

Terraform Task-2

Task Description:

Create 2 EC2 instances on 2 different regions and install nginx using terraform script.

Step 1 : Configured aws cli in terraform server.

```
ubuntu@ip-172-31-40-191:~$ aws configure
AWS Access Key ID [None]: AKIA4F35FI6UL3H0DB60
AWS Secret Access Key [None]: /5Hoewbtd3FtkUR/vLbMkmICmkM622qckkwhYVbJ
Default region name [None]: us-west-2
Default output format [None]:
```

Step 3 - Executed the script

terraform init

```
ubuntu@ip-172-31-40-191:~/project2$ terraform init
Initializing the backend...
Initializing provider plugins...
- Finding hashicorp/aws versions matching "~> 5.0"...
- Installing hashicorp/aws v5.100.0...
- Installed hashicorp/aws v5.100.0 (signed by HashiCorp)
Terraform has created a lock file .terraform.lock.hcl to record the provider
selections it made above. Include this file in your version control repository
so that Terraform can guarantee to make the same selections by default when
you run "terraform init" in the future.

Terraform has been successfully initialized!
```

terraform apply

```
ubuntu@ip-172-31-40-191:~/project2$ terraform apply
data.aws_ami.al2023_us: Reading...
data.aws_ami.al2023_in: Reading...
data.aws_ami.al2023_us: Read complete after 1s [id=ami-02f058bc6b3ee6fcc]
data.aws_ami.al2023_in: Read complete after 2s [id=ami-02a8bcaaf6d87b759]

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
+ create

Terraform will perform the following actions:
```

Instances started running in two regions

Instance summary for i-01077ae44608eb7bd (nginx-us-east-1) [Info](#)

Updated less than a minute ago

Instance ID

[i-01077ae44608eb7bd](#)

IPV6 address

–

Hostname type

IP name: ip-172-31-21-100.ec2.internal

Answer private resource DNS name

–

Auto-assigned IP address

[52.91.170.104](#) [Public IP]

IAM role

–

IMDSv2

Required

Operator

Public IPv4 address

[52.91.170.104](#) | [open address](#)

Instance state

[Running](#)

Private IP DNS name (IPv4 only)

[ip-172-31-21-100.ec2.internal](#)

Instance type

[t3.micro](#)

VPC ID

[vpc-03e1852553eb9dd75](#)

Subnet ID

[subnet-02e8cc4a115b24ff6](#)

Instance ARN

[arn:aws:ec2:us-east-1:837244241832:instance/i-01077ae44608eb7bd](#)

Private IPv4 addresses

[172.31.21.100](#)

Public DNS

[ec2-52-91-170-104.compute-1.amazonaws.com](#) | [open address](#)

Elastic IP addresses

–

AWS Compute Optimizer finding

[Opt-in to AWS Compute Optimizer for recommendations.](#) | [Learn more](#)

Auto Scaling Group name

–

Managed

false

[⚠ Not secure](#) | [52.91.170.104](#)

Welcome to nginx!

If you see this page, the nginx web server is successfully installed and working. Further configuration is required.

For online documentation and support please refer to [nginx.org](#).
Commercial support is available at [nginx.com](#).

Thank you for using nginx.

Instance summary for i-0348ead604e4d5687 (nginx-ap-south-1) Info

Refreshing instance data

Instance ID
i-0348ead604e4d5687

IPv6 address
-

Hostname type
IP name: ip-172-31-44-115.ap-south-1.compute.internal

Answer private resource DNS name
-

Auto-assigned IP address
C

IAM role
-

IMDSv2
Required

Operator
-

Public IPv4 address
13.232.130.244 | [open address](#)

Instance state
Running

Private IP DNS name (IPv4 only)
ip-172-31-44-115.ap-south-1.compute.internal

Instance type
t3.micro

VPC ID
vpc-00f1b106cd708742 | [open address](#)

Subnet ID
subnet-003ad8b0f91fd320d | [open address](#)

Instance ARN
arn:aws:ec2:ap-south-1:837244241832:instance/i-0348ead604e4d5687

Private IPv4 addresses
172.31.44.115

Public DNS
ec2-13-232-130-244.ap-south-1.compute.amazonaws.com | [open address](#)

Elastic IP addresses
C

AWS Compute Optimizer finding
Opt-in to AWS Compute Optimizer for recommendations. | [Learn more](#)

Auto Scaling Group name
-

Managed
false

Not secure | 13.232.130.244

Welcome to nginx!

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Thank you for using nginx.

Step 2 - Created a terraform script to launch instances in N. Verginia and Mumbai

```

terraform {

  required_providers {

    aws = {

      source = "hashicorp/aws"

      version = "~> 5.0"

    }

  }

}

provider "aws" {

```

```
alias = "us"

region = "us-east-1"

}

provider "aws" {

  alias = "in"

  region = "ap-south-1"

}

data "aws_ami" "al2023_us" {

  provider = aws.us

  most_recent = true

  owners = ["amazon"]

  filter {

    name = "name"

    values = ["al2023-ami-*-x86_64"]

  }

}

data "aws_ami" "al2023_in" {

  provider = aws.in

  most_recent = true

  owners = ["amazon"]

  filter {

    name = "name"

    values = ["al2023-ami-*-x86_64"]

  }

}

resource "aws_security_group" "web_us" {
```

```
provider = aws.us
```

```
name     = "nginx-sg-us"
```

```
ingress {
```

```
  from_port = 22
```

```
  to_port   = 22
```

```
  protocol  = "tcp"
```

```
  cidr_blocks = ["0.0.0.0/0"]
```

```
}
```

```
ingress {
```

```
  from_port = 80
```

```
  to_port   = 80
```

```
  protocol  = "tcp"
```

```
  cidr_blocks = ["0.0.0.0/0"]
```

```
}
```

```
egress {
```

```
  from_port = 0
```

```
  to_port   = 0
```

```
  protocol  = "-1"
```

```
  cidr_blocks = ["0.0.0.0/0"]
```

```
}
```

```
}
```

```
resource "aws_security_group" "web_in" {
```

```
  provider = aws.in
```

```
  name     = "nginx-sg-in"
```

```
  ingress {
```

```
    from_port = 22
```

```
    to_port    = 22

    protocol   = "tcp"

    cidr_blocks = ["0.0.0.0/0"]
}

ingress {

    from_port = 80

    to_port   = 80

    protocol   = "tcp"

    cidr_blocks = ["0.0.0.0/0"]
}

egress {

    from_port = 0

    to_port   = 0

    protocol   = "-1"

    cidr_blocks = ["0.0.0.0/0"]
}
}

resource "aws_instance" "nginx_us" {

    provider    = aws.us

    ami         = data.aws_ami.al2023_us.id

    instance_type = "t3.micro"

    security_groups = [

        aws_security_group.web_us.name
    ]

    user_data = <<-EOF

    #!/bin/bash
```

```
dnf update -y
```

```
dnf install -y nginx
```

```
systemctl enable nginx
```

```
systemctl start nginx
```

```
EOF
```

```
tags = {
```

```
    Name = "nginx-us-east-1"
```

```
}
```

```
}
```

```
resource "aws_instance" "nginx_in" {
```

```
    provider    = aws.in
```

```
    ami         = data.aws_ami.al2023_in.id
```

```
    instance_type = "t3.micro"
```

```
    security_groups = [
```

```
        aws_security_group.web_in.name
```

```
    ]
```

```
    user_data = <<-EOF
```

```
        #!/bin/bash
```

```
        dnf update -y
```

```
        dnf install -y nginx
```

```
        systemctl enable nginx
```

```
        systemctl start nginx
```

```
EOF
```

```
tags = {
```

```
    Name = "nginx-ap-south-1"
```

```
}
```

}